Dictionary-Linked Tucker Models for Nonlinear Low-Rank Tensor Signal Recovery

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***Abstract*—Many multidimensional signals are compactly represented by low-rank tensor models, but nonlinear measurement responses can make a low-rank latent signal appear higher-rank in the observed domain. This paper studies a dictionary-linked Tucker model for nonlinear low-rank tensor signal recovery. The model keeps a fixed Tucker backbone and learns a low-dimensional scalar response jointly with the latent tensor, yielding a shared-backbone rank lift with few additional parameters. The finite-dictionary formulation allows different scalar response bases; this paper uses the simplest power dictionary, which yields the polynomial-linked model and aligns directly with the rank-lift analysis. We analyze the induced rank expansion, response approximation, stationarity of the alternating estimator, and computational scaling. We also develop a frozen-backbone link diagnostic that predicts when the nonlinear response is likely to help. Experiments on controlled nonlinear tensors, hyperspectral reconstruction and completion, noisy reconstruction, and cross-domain diagnostics show that the link improves recovery when the residual is coherent with a shared measurement response, while low-diagnostic regimes clarify the operating conditions of the model.**

***Index Terms*—Tensor decomposition, Tucker decomposition, low-rank tensor recovery, nonlinear measurements, tensor completion, hyperspectral imaging.**

I. Introduction

MULTIDIMENSIONAL signals often contain strong structure across space, spectrum, time, view, or acquisition mode. Low-rank tensor models exploit this structure by representing a signal through a small number of coupled latent factors, and have become standard tools for signal representation, denoising, completion, and recovery [1]–[4]. Tucker decomposition is especially useful in this setting because it separates a compact core from mode-wise factors and provides an explicit multilinear rank budget. In the classical Tucker model, however, the reconstructed tensor is fitted directly to the measured tensor. This amounts to assuming that the map from the latent low-rank signal to the observed measurements is linear.

This linear measurement assumption is often an idealization. Imaging and sensing pipelines may include camera response curves, radiometric calibration effects, saturation, reflectance nonlinearities, or other response mechanisms between the latent physical quantity and the stored measurement [5]–[8]. When such a response acts entrywise on a low-rank latent tensor, the measured tensor need not remain low-rank in the same Tucker sense. A low-rank latent signal can therefore appear to require a substantially larger multilinear rank after measurement. From a signal recovery perspective, this creates a model mismatch: the rank budget selected for the latent signal may be appropriate, while the linear Tucker observation model is not.

One way to reduce this mismatch is to increase the Tucker rank. This is often effective, but it changes the entire multilinear representation: the core grows rapidly, each mode factor gains additional degrees of freedom, and the interpretation of a fixed latent rank budget is lost. A second approach is to replace the multilinear model with a more flexible nonlinear tensor model, such as a kernel, Gaussian-process, generalized-link, or neural tensor decoder [9]–[11]. Such methods are powerful, but their flexibility can obscure the relation between the recovered signal and a chosen Tucker backbone, and may introduce additional training and model-selection burdens.

This paper studies a narrower and more structured alternative. We assume that the latent signal is well represented by a Tucker tensor, but that the measurements are generated from that latent tensor through an unknown scalar response. We approximate this response in a finite scalar dictionary and learn the response jointly with the Tucker backbone. If

$$\mathcal{S} = [\![\mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)}]\!] \tag{1}$$

is the latent Tucker signal, the measured tensor is modeled as

$$\widehat{\mathcal{Y}} = f_{\boldsymbol{\beta}}(\mathcal{S}), \tag{2}$$

where $f_{\boldsymbol{\beta}}(s) = \sum_{q=0}^{Q} \beta_q \psi_q(s)$ for a chosen scalar dictionary $\{\psi_q\}_{q=0}^{Q}$. We call this a dictionary-linked Tucker model. It keeps the Tucker backbone ranks fixed and adds only the scalar link coefficients $\boldsymbol{\beta} = (\beta_0, \dots, \beta_Q)$. Different finite scalar dictionaries can be used in this formulation when response priors are available. In this paper, we use the simplest power dictionary $\psi_q(s) = s^q$, which gives the polynomial-linked model and makes the rank-lift mechanism explicit. The model is therefore not an unrestricted nonlinear decoder and not a post-hoc calibration of a fixed Tucker fit. The latent tensor and the response are estimated together, so the Tucker backbone is shaped by the nonlinear measurement model during recovery.

The key structural point is most explicit for the polynomial dictionary, where the link creates a shared-backbone rank lift. The degree-p term $\mathcal{S}^{\odot p}$ can be viewed as a Tucker tensor whose factors are polynomial feature lifts of the original factors. Thus higher-degree channels can increase the effective multilinear rank of the measured tensor, but they remain tied to one latent Tucker backbone. More general dictionaries keep the same modeling principle: the additional expressivity must be explainable by a shared scalar response acting on the latent signal. This constraint is precisely what makes the model useful in the intended regime and limited outside it. If the

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remaining residual is ordinary multilinear structure, increasing Tucker rank is the appropriate remedy; if the residual is coherent with a measurement response, a scalar link can provide a compact correction.

We develop this idea for masked tensor signal recovery, covering both full reconstruction and tensor completion. The estimation procedure alternates between a small ridge-regression update for the scalar link and gradient-based updates of the Tucker backbone. To improve numerical stability, the implementation uses normalized features and continuation when the selected dictionary is nested. We also introduce a frozen-backbone link diagnostic that measures how much residual energy can be removed by one link update after a linear Tucker fit. This diagnostic is designed to answer a practical question before full nonlinear training: does this signal contain residual structure that is likely to benefit from a shared scalar response?

The resulting study is organized around the full signal-recovery pipeline: representation, estimation, diagnosis, and validation. On the representation side, we characterize when a scalar response creates an efficient rank lift and when ordinary Tucker rank expansion is the more direct explanation. On the estimation side, we use the special structure of the link to keep the nonlinear part low-dimensional. On the experimental side, we evaluate both high- and low-diagnostic regimes to clarify the operating conditions of the model.

A. Contributions

The main contributions are summarized as follows.

- We formulate nonlinear low-rank tensor recovery under an unknown entrywise measurement response, separating the rank of the latent signal from the rank inflation observed after measurement.
- We propose a dictionary-linked Tucker model that keeps a fixed Tucker backbone while learning a low-dimensional scalar response. The estimator is instantiated with a power dictionary, yielding a compact polynomial link tied to the same latent backbone.
- We analyze the resulting model class and estimation procedure, including induced rank expansion, parameter economy relative to explicit lifted Tucker models, response approximation, descent properties, diagnostic interpretation, and computational scaling.
- We evaluate the model on controlled nonlinear tensor signals and hyperspectral recovery tasks, with rank-lift comparisons, noisy-observation robustness tests, and diagnostic stratification that identifies when the nonlinear response is useful.

The remainder of the paper develops the model, analysis, estimation algorithm, diagnostic score, and experimental validation in this order.

II. Related Work

A. Low-Rank Tensor Signal Models

Low-rank tensor decompositions are a standard language for multidimensional signal modeling. CP and Tucker models have been used for compression, denoising, source separation, completion, and data fusion across signal processing and machine learning [1], [2], [12]. Tucker decomposition, in particular, represents a tensor by a small core and mode-wise factors [13], making the multilinear rank explicit. Other structured formats, including tensor-train and tensor-ring decompositions [14], [15], control complexity through different network topologies. The present paper uses Tucker as the backbone because its mode-wise ranks give a direct way to distinguish latent signal rank from the effective rank of the measured tensor.

Tensor completion methods exploit these low-dimensional structures to recover missing entries. Convex and nonconvex approaches include nuclear-norm inspired formulations, factorized Tucker or CP objectives, and practical algorithms for incomplete tensors [3], [4], [16]. These methods usually model the measured tensor itself as low-rank. Our setting is different: the latent signal is low-rank, but an entrywise response can lift the measured tensor outside the same rank budget. Increasing Tucker rank can reduce the mismatch, but doing so changes every mode factor and grows the core. NTD-PL instead keeps the original backbone rank and adds a low-dimensional response tied to that backbone.

B. Generalized and Nonlinear Tensor Models

Several tensor models replace the Gaussian squared-loss view with generalized or nonlinear observation models. Generalized CP and probabilistic tensor factorizations model non-Gaussian observations through a chosen likelihood or link function [17], [18], while nonparametric Bayesian tensor models increase flexibility through latent function priors [19]. Kernel, functional, and neural tensor methods further expand the model class by learning nonlinear feature maps or deep decoders [9]–[11], [20]. These approaches are valuable when the data require broad nonlinear expressivity, but they can make it harder to interpret how a fixed multilinear rank budget is being used.

The dictionary-linked Tucker model uses a constrained scalar response. It assumes that the dominant residual left by a fixed Tucker backbone can be explained by a shared entrywise map, while preserving the interpretability of the Tucker ranks. This places the model between classical low-rank recovery and general nonlinear tensor factorization. The power-dictionary instance is expressive enough to approximate common smooth responses on a bounded signal range, while remaining structured enough to admit a rank-expansion analysis and a closed-form link update. Other finite scalar dictionaries could be substituted when calibration constraints suggest a different response basis, without replacing the Tucker backbone.

C. Imaging Recovery and Nonlinear Responses

Nonlinear responses arise naturally in imaging and sensing pipelines. Camera response curves, radiometric processing, saturation, and high-dynamic-range imaging motivate explicit response modeling in computational photography [5], [6]. Hyperspectral imaging adds additional sources of nonlinear behavior, including nonlinear mixing, reflectance effects, and sensor-dependent preprocessing [7], [8]. At the same time, hyperspectral cubes are strongly correlated across spatial and spectral modes, making them natural testbeds for low-rank tensor recovery.

Hyperspectral reconstruction and completion provide a signal recovery setting where fixed-rank Tucker mismatch is observable and measurable. The proposed link diagnostic estimates, before full nonlinear training, whether a frozen Tucker residual contains structure that a scalar response can remove. This makes the model selection criterion explicit and testable.

III. Dictionary-Linked Tucker Representation

We first describe the nonlinear measurement model and then introduce the dictionary-linked Tucker representation used for recovery. The goal is not to define a new tensor task, but to make explicit a common source of model mismatch: the tensor that is compact in the latent signal domain may be observed after a nonlinear response.

A. Nonlinear Low-Rank Signal Model

Let $\mathcal{Y} \in \mathbb{R}^{I_1\times\cdots\times I_N}$ denote the measured tensor and let $\Omega \in \{0,1\}^{I_1\times\cdots\times I_N}$ be the observation mask. We assume that the measured entries are generated from a latent tensor $\mathcal{S}$ through a scalar response,

$$Y_{\mathbf{i}} = f(S_{\mathbf{i}}) + \varepsilon_{\mathbf{i}}, \qquad \mathbf{i} = (i_1,\ldots,i_N), \tag{3}$$

where $\varepsilon_{\mathbf{i}}$ denotes measurement noise. The latent tensor is represented by a Tucker model,

$$\mathcal{S} = [\![\mathcal{X}; \mathbf{A}^{(1)},\ldots,\mathbf{A}^{(N)}]\!], \tag{4}$$

where $\mathcal{X} \in \mathbb{R}^{R_1\times\cdots\times R_N}$ is the core and $\mathbf{A}^{(n)} \in \mathbb{R}^{I_n\times R_n}$ are mode factors.

This model separates two notions of complexity. The latent signal is controlled by the Tucker rank $(R_1,\ldots,R_N)$, while the measured tensor may have larger effective multilinear rank because of the response f. The recovery problem is therefore to estimate the low-rank latent structure and the response from the observed entries of $\mathcal{Y}$.

B. Scalar Link Dictionaries

We approximate the unknown response in (3) by a finite scalar dictionary,

$$\widehat{\mathcal{Y}} = f_{\boldsymbol\beta}(\mathcal{S}) = \sum_{q=0}^{Q} \beta_q \psi_q(\mathcal{S}), \tag{5}$$

where $\psi_q(\mathcal{S})$ applies ψ_q entrywise. Standard Tucker is recovered whenever the dictionary contains the identity channel and the fitted response is $f_{\boldsymbol\beta}(s) = s$. The default choice is the power dictionary $\psi_q(s) = s^q$, which gives the polynomial-linked Tucker model

$$f_{\boldsymbol\beta}(\mathcal{S}) = \sum_{p=0}^{P} \beta_p \mathcal{S}^{\odot p}. \tag{6}$$

For $p \geq 2$, the degree-p channel $\mathcal{S}^{\odot p}$ corresponds to a tied high-order interaction among copies of the same Tucker backbone. Thus the model can increase effective multilinear rank without adding an independent high-rank tensor block. Appendix A-A gives the explicit p-fold Tucker construction from which this tied channel is obtained.

The same estimator can also support non-power dictionaries that remain linear in the coefficients $\boldsymbol\beta$, for example conditioned polynomial bases, radial basis functions, or splines. These choices change the scalar response family but not the Tucker backbone, the masked objective, or the link-refresh subproblem. Unless otherwise stated, the remainder of the paper uses the power dictionary.

C. Masked Recovery Objective

The same formulation covers full reconstruction and completion. We estimate the core, factors, and link coefficients by minimizing

$$\min_{\mathcal{X},\{\mathbf{A}^{(n)}\},\boldsymbol\beta} \frac{1}{2|\Omega|} \|\Omega \odot (\mathcal{Y} - f_{\boldsymbol\beta}(\mathcal{S}))\|_F^2 + \mathcal{R}(\mathcal{X}, \{\mathbf{A}^{(n)}\}, \boldsymbol\beta), \tag{7}$$

where $\mathcal{R}$ contains regularization terms for the Tucker backbone and the scalar link. When Ω is the all-one tensor, (7) gives full reconstruction. When Ω selects a subset of entries, the loss is evaluated only on observed measurements and prediction is assessed on the held-out entries.

IV. Estimation Algorithm and Link Diagnostic

Let $m = |\Omega|$ and let $\theta = (\mathcal{X}, \{\mathbf{A}^{(n)}\}_{n=1}^N, \boldsymbol\beta)$ collect all trainable parameters. For an active link dictionary of size $Q+1$, we minimize

$$\begin{aligned} \mathcal{F}_Q(\theta) = \frac{1}{2|\Omega|} \|\Omega \odot (\mathcal{Y} - f_{\boldsymbol\beta}(\mathcal{S}))\|_F^2 + \frac{\lambda_X}{2}\|\mathcal{X}\|_F^2 \\ + \sum_{n=1}^{N} \frac{\lambda_n}{2}\left\|\mathbf{A}^{(n)}\right\|_F^2 + \frac{\lambda_\beta}{2}\|\boldsymbol\beta\|_2^2, \end{aligned} \tag{8}$$

where $f_{\boldsymbol\beta}(s) = \sum_{q=0}^{Q} \beta_q \psi_q(s)$ and $\{\psi_q\}_{q=0}^{Q}$ is the selected scalar link dictionary. The monomial basis $\psi_q(s) = s^q$ gives (6); in implementation, the same polynomial space may be represented by centered, Chebyshev, or otherwise conditioned bases. Non-polynomial choices such as RBF or spline dictionaries use the same objective as long as the response remains linear in $\boldsymbol\beta$. The experiments use the monomial power dictionary. The estimation alternates between an exact low-dimensional link update and a descent step for the Tucker backbone.

A. Link Refresh as a Ridge Subproblem

Fix the latent tensor $\mathcal{S}$ and list the observed entries as $y_\Omega \in \mathbb{R}^m$ and $s_\Omega \in \mathbb{R}^m$. Define the feature matrix $\mathbf{\Phi}_{\Omega,Q}(\mathcal{S}) \in \mathbb{R}^{m\times(Q+1)}$ by

$$[\mathbf{\Phi}_{\Omega,Q}(\mathcal{S})]_{j,q} = \psi_q([s_\Omega]_j), \qquad q = 0,\ldots,Q. \tag{9}$$

The link update is the ridge regression

$$\boldsymbol\beta^+ = \arg\min_{\boldsymbol\beta\in\mathbb{R}^{Q+1}} \frac{1}{2m}\|y_\Omega - \mathbf{\Phi}_{\Omega,Q}(\mathcal{S})\boldsymbol\beta\|_2^2 + \frac{\lambda_\beta}{2}\|\boldsymbol\beta\|_2^2, \tag{10}$$

with closed-form solution

$$\boldsymbol{\beta}^{+}=\left(\frac{1}{m}\boldsymbol{\Phi}_{\Omega,Q}^{T}\boldsymbol{\Phi}_{\Omega,Q}+\lambda_{\beta}\mathbf{I}\right)^{-1}\frac{1}{m}\boldsymbol{\Phi}_{\Omega,Q}^{T}y_{\Omega}. \tag{11}$$

This step solves only a $(Q+1)\times(Q+1)$ system. It is therefore cheap relative to reconstructing or differentiating through the Tucker tensor, and because it solves the $\boldsymbol{\beta}$-subproblem exactly, it never increases $\mathcal{F}_Q$ for the current backbone.

B. Linked-Backbone Descent

With $\boldsymbol{\beta}$ fixed, the residual is backpropagated through the scalar response and then through the Tucker map. The gradient with respect to the latent tensor is

$$\mathbf{G}_{\mathcal{S}}=\frac{1}{m}\Omega\odot(f_{\boldsymbol{\beta}}(\mathcal{S})-\mathcal{Y})\odot f_{\boldsymbol{\beta}}'(\mathcal{S}). \tag{12}$$

The Tucker gradients are standard reverse contractions:

$$\nabla_{\mathcal{X}}\mathcal{F}_Q=\mathbf{G}_{\mathcal{S}}\times_1(\mathbf{A}^{(1)})^T\cdots\times_N(\mathbf{A}^{(N)})^T+\lambda_X\mathcal{X}, \tag{13}$$

$$\nabla_{\mathbf{A}^{(n)}}\mathcal{F}_Q=[\mathbf{G}_{\mathcal{S}}]_{(n)}\mathbf{Z}_n^T+\lambda_n\mathbf{A}^{(n)}, \tag{14}$$

where $\mathbf{Z}_n=(\mathbf{A}^{(N)}\otimes\cdots\otimes\mathbf{A}^{(n+1)}\otimes\mathbf{A}^{(n-1)}\otimes\cdots\otimes\mathbf{A}^{(1)})\mathcal{X}_{(n)}^T$. The backbone update can use a line-search gradient step, a block coordinate gradient step, or an adaptive first-order implementation, provided the step satisfies the sufficient decrease condition used in Section V.

C. Dictionary Continuation and Normalization

Large or high-order link dictionaries can amplify Tucker scale ambiguity. For nested dictionaries, such as the power or Chebyshev bases, we use continuation: training starts with the Tucker-compatible identity response and activates additional channels gradually. When a new channel is introduced, its coefficient is initialized at zero, so the prediction is unchanged at the moment of expansion. Non-nested dictionaries such as RBF and spline bases are trained with the full selected dictionary from the start. After backbone updates, factor columns are rescaled and the inverse scales are absorbed into the core. This fixes a numerical gauge without changing the latent tensor $\mathcal{S}$ or the objective value.

D. Alternating Recovery Scheme

One outer iteration consists of:

1) fixing the current Tucker backbone and refreshing $\boldsymbol{\beta}$ by (11);
2) updating $\mathcal{X}$ and $\{\mathbf{A}^{(n)}\}$ by a descent step using (12)–(14);
3) normalizing the Tucker gauge and, when the dictionary is nested, increasing the active dictionary size according to a continuation schedule.

This procedure treats the link as part of the fitted representation: the backbone update is computed through the current nonlinear measurement map.

E. Link Diagnostic

The link diagnostic measures whether the residual of a linear Tucker fit is coherent with a scalar response. Let $\mathcal{S}_T$ be a fitted Tucker reconstruction and let $\boldsymbol{\beta}_T=(0,1,0,\ldots,0)$ denote the identity link. The frozen-backbone diagnostic is

$$E_{\text{lin}}=\|y_\Omega-\boldsymbol{\Phi}_{\Omega,1}(\mathcal{S}_T)\boldsymbol{\beta}_T\|_2^2, \tag{15}$$

$$E_{\text{link}}=\min_{\boldsymbol{\beta}}\left[\|y_\Omega-\boldsymbol{\Phi}_{\Omega,Q}(\mathcal{S}_T)\boldsymbol{\beta}\|_2^2+m\lambda_\beta\|\boldsymbol{\beta}\|_2^2\right], \tag{16}$$

$$D_{\text{link}}=\frac{E_{\text{lin}}-E_{\text{link}}}{E_{\text{lin}}}. \tag{17}$$

Large D_{link} means that a low-dimensional scalar response can already remove substantial residual energy without changing the Tucker backbone. We use this score as a model-selection and interpretation tool: high diagnostic yield suggests that the nonlinear link is aligned with the residual structure, whereas low diagnostic yield suggests that increasing multilinear rank or changing the tensor prior may be more appropriate.

V. Analysis

This section summarizes the structural, optimization, and computational properties of dictionary-linked Tucker models. The rank results are stated for the power dictionary, where the shared-backbone lift has an explicit algebraic form. The optimization results apply to any differentiable finite dictionary whose response is linear in the link coefficients.

Proposition 1 (Shared-backbone rank expansion)**.** *Let $\mathcal{S}$ have Tucker rank at most $(R_1,\ldots,R_N)$. For every $p\geq 1$, $\mathcal{S}^{\odot p}$ admits a Tucker representation with mode ranks at most*

$$\left(\binom{R_1+p-1}{p},\ldots,\binom{R_N+p-1}{p}\right). \tag{18}$$

Thus each polynomial channel can be represented as an ordinary Tucker tensor with lifted factors, but those lifted factors remain tied to the same original backbone. This is the structural difference between the proposed model and an untied higher-rank correction.

Corollary 1 (Parameter economy of a tied lift)**.** *Let $D_n(p)=\binom{R_n+p-1}{p}$. A Tucker model that explicitly represents the degree-p channel in Proposition 1 may require*

$$d_{\text{lift}}(p)=\prod_{n=1}^{N}D_n(p)+\sum_{n=1}^{N}I_nD_n(p) \tag{19}$$

parameters for that channel alone. The power-dictionary linked model represents all degrees $0,\ldots,Q$ with the original Tucker parameters plus $Q+1$ link coefficients. Hence the extra parameter cost of the linked model is independent of the ambient dimensions $\{I_n\}$ and the lifted ranks $\{D_n(p)\}$.

This corollary does not imply that an explicit higher-rank Tucker model is a poor model; it explains why the linked model is economical when the high-rank structure is specifically generated by a shared scalar response.

Theorem 1 (Separation from matched-rank Tucker)**.** *Let $N\geq 3$, $R\geq 2$, and $p\geq 2$. Define $D=\binom{R+p-1}{p}$ and suppose $I_n\geq D$ for every mode. Then there exists an order-N tensor* *representable by a degree-p polynomial-linked Tucker model with backbone rank $(R,\dots,R)$, but whose exact standard Tucker rank is $(D,\dots,D)$.*

The theorem gives a strict algebraic separation: the power link does more than post-calibrate an already fitted tensor. It can generate tensors whose ordinary multilinear ranks match the Veronese-lifted ranks in Proposition 1, while the trainable backbone remains at rank $(R,\dots,R)$. This is an existence result for shared-response high-rank structure, not a claim that the linked model spans all tensors of the larger Tucker rank.

Theorem 2 (Polynomial response generation)**.** *If $\mathcal{Y} = g(\mathcal{S}^\star)$, where $\mathcal{S}^\star$ has Tucker rank at most $(R_1,\dots,R_N)$ and g is a polynomial of degree at most P, then a degree-P polynomial-linked Tucker model with the same backbone rank can represent $\mathcal{Y}$ exactly.*

Proposition 2 (Continuous response approximation)**.** *Let $\mathcal{S}^\star$ have Tucker rank at most $(R_1,\dots,R_N)$ and entries in a compact interval $[a,b]$. If the measurement response f is continuous on $[a,b]$, then for every $\epsilon > 0$ there exists a polynomial degree P and coefficients $\boldsymbol{\beta}$ such that*

$$\|f(\mathcal{S}^\star) - f_{\boldsymbol{\beta}}(\mathcal{S}^\star)\|_F \le \epsilon\sqrt{M}, \qquad M = \prod_{n=1}^{N} I_n. \tag{20}$$

Thus the power link is not restricted to exactly polynomial sensors. On a bounded signal range, it can approximate any smooth or continuous scalar response to the desired uniform accuracy, while keeping the latent Tucker rank fixed. The degree required for a given accuracy depends on the regularity of the response and the width of the signal range. This approximation result is the reason the power dictionary is used as the default scalar basis in the experiments.

Remark 1 (Gauge and identifiability)**.** *As in ordinary Tucker models, the factorization of $\mathcal{S}$ is not unique: mode factors can be rescaled or transformed with an inverse transformation absorbed by the core. The scalar link adds another scale interaction because changing the scale of $\mathcal{S}$ changes the natural scale of the link coefficients. The estimator therefore fixes a numerical gauge by normalizing factor columns and absorbing inverse scales into the core. This does not make the Tucker factors identifiable in a strict statistical sense, but it keeps the represented latent tensor and response numerically well conditioned and makes the objective in Section IV a stable parameterization of the same fitted measurement tensor.*

A. Descent and Stationarity

We analyze the fixed-dictionary objective $\mathcal{F}_Q$ in (8). For nested dictionaries, the continuation schedule can be viewed as a sequence of such fixed-dictionary problems, each initialized from the previous dictionary size.

Assumption 1. *For a fixed active dictionary size $Q+1$, the iterates remain in a bounded normalized parameter set. The regularization weights are nonnegative and $\lambda_\beta > 0$. The backbone step either uses a line search or an equivalent descent rule such that, for constants $c > 0$ and $b > 0$,*

$$\mathcal{F}_Q(\theta^k) - \mathcal{F}_Q(\theta^{k+1}) \ge c\left\|\theta^{k+1} - \theta^k\right\|_2^2, \tag{21}$$

$$\operatorname{dist}\bigl(0, \partial\mathcal{F}_Q(\theta^{k+1})\bigr) \le b\left\|\theta^{k+1} - \theta^k\right\|_2. \tag{22}$$

The boundedness condition is enforced in practice by Tucker gauge normalization and regularization. The two descent conditions are standard for block coordinate and line-search gradient schemes on smooth objectives.

Proposition 3 (Monotone objective decrease)**.** *Under Assumption 1, every fixed-dictionary outer iteration of the alternating estimator satisfies*

$$\mathcal{F}_Q(\theta^{k+1}) \le \mathcal{F}_Q(\theta^k). \tag{23}$$

Moreover, if (21) *holds for the full outer update, then*

$$\sum_{k=0}^{\infty}\left\|\theta^{k+1} - \theta^k\right\|_2^2 < \infty. \tag{24}$$

Theorem 3 (Convergence to a stationary point)**.** *Suppose Assumption 1 holds for a fixed dictionary size $Q+1$. Then the sequence generated by the alternating estimator has finite length and converges to a critical point $\theta^\star$ of $\mathcal{F}_Q$:*

$$0 \in \partial\mathcal{F}_Q(\theta^\star). \tag{25}$$

The theorem does not claim global optimality; the problem remains nonconvex because of the Tucker backbone. Its role is to rule out pathological cycling for the proposed alternating estimator and to place the algorithm on the same theoretical footing as standard nonconvex block-descent methods [21], [22].

B. Diagnostic Interpretation

The diagnostic D_{link} in (17) can be interpreted as a projection test for link-like residual structure. Let $\mathbf{\Phi}_{\Omega,Q}$ be the scalar-link feature matrix built from a frozen Tucker reconstruction $\mathcal{S}_T$ and let $\mathbf{P}_Q$ denote the orthogonal projector onto its column space. When $\lambda_\beta = 0$, the refreshed-link residual satisfies

$$E_{\text{link}} = \|(\mathbf{I} - \mathbf{P}_Q)y_\Omega\|_2^2. \tag{26}$$

Consequently, D_{link} is the fraction of identity-link residual energy removed by projecting the observations onto the scalar-response subspace generated by the frozen backbone. With ridge regularization, the same interpretation holds with $\mathbf{P}_Q$ replaced by the corresponding shrinkage operator. Large D_{link} therefore means that the residual is aligned with scalar functions of the current latent estimate, whereas small D_{link} suggests that the residual is better addressed by increasing multilinear rank, changing the tensor prior, or revisiting the observation model.

C. Masked Generalization

The masked objective also has the usual sampling interpretation. Let $\Theta \subset \mathbb{R}^d$ be a bounded parameter set for a degree-P polynomial-linked Tucker model, where

$$d = \prod_{n=1}^{N} R_n + \sum_{n=1}^{N} I_n R_n + (P+1). \tag{27}$$

For $\theta \in \Theta$, define the full-entry risk $\mathcal{L}(\theta) = M^{-1} \sum_{\mathbf{i}} \ell_{\mathbf{i}}(\theta)$, $M = \prod_n I_n$, and the masked empirical risk $\widehat{\mathcal{L}}_m(\theta) = m^{-1} \sum_{t=1}^{m} \ell_{\mathbf{i}_t}(\theta)$, where the entries $\mathbf{i}_t$ are sampled uniformly.

Theorem 4 (Masked-entry uniform convergence)**.** *Assume* $0 \le \ell_{\mathbf{i}}(\theta) \le B_\ell$ *and* $\ell_{\mathbf{i}}(\theta)$ *is G-Lipschitz in* θ*, uniformly over* $\mathbf{i}$*. If* Θ *lies in a Euclidean ball of radius* B_Θ*, then, with probability at least* $1-\delta$*,*

$$\sup_{\theta \in \Theta} \left| \mathcal{L}(\theta) - \widehat{\mathcal{L}}_m(\theta) \right| \le 2G\varepsilon + B_\ell \sqrt{\frac{d \log(3B_\Theta/\varepsilon) + \log(2/\delta)}{2m}} \tag{28}$$

for every $0 < \varepsilon \le B_\Theta$*.*

This bound is not intended as a sharp sample-complexity result. Its role is to make clear which complexity enters the masked recovery problem: the covering dimension is the tied dictionary-linked parameter count, rather than the parameter count of an untied Veronese-rank Tucker expansion.

D. Computational Scaling

Let $M = \prod_{n=1}^{N} I_n$, $R_\Pi = \prod_{n=1}^{N} R_n$, and $d_{\text{Tucker}} = R_\Pi + \sum_{n=1}^{N} I_n R_n$. Storing the dictionary-linked model requires

$$d_{\text{NTD}} = d_{\text{Tucker}} + (Q+1) \tag{29}$$

parameters. A rank-expanded Tucker model that explicitly matches the degree-Q Veronese lift may require mode ranks $D_n = \binom{R_n+Q-1}{Q}$ for the degree-Q channel, with core size $\prod_n D_n$ and factor size $\sum_n I_n D_n$. The proposed model keeps the storage cost tied to the original backbone and avoids explicit storage of the lifted tensor.

For one outer iteration, Link Refresh forms $\mathbf{\Phi}_{\Omega,Q}^T \mathbf{\Phi}_{\Omega,Q}$ and solves a small system, costing

$$\mathcal{O}\big(mQ^2 + Q^3\big) \, . \tag{30}$$

The dominant backbone cost is evaluating the Tucker reconstruction and reverse contractions on the observed set or on the full tensor, depending on the implementation. Dense reconstruction scales as

$$\mathcal{O}\left(M R_\Pi + \sum_{n=1}^{N} I_n R_n R_\Pi / R_n \right) \tag{31}$$

up to constants determined by contraction order, followed by an additional $\mathcal{O}(MQ)$ cost to evaluate the scalar response for dense dictionaries. In sparse masked settings, the same operations can be restricted to observed entries, replacing M by m in the response and residual computations. Since Q is small, the link cost is typically negligible compared with the Tucker contractions.

VI. Experiments

The experiments evaluate the proposed model through three connected questions. First, does the scalar link behave as predicted on controlled nonlinear measurements? Second, under the same Tucker backbone rank, does the linked response improve real tensor reconstruction and completion consistently across scenes, with separate stress tests for missing entries and observation noise? Third, can a low-cost diagnostic indicate when the link is appropriate? These questions match the modeling claim: NTD-PL is expected to be most beneficial when a matched-rank Tucker model leaves residual structure consistent with a shared scalar response. Unless stated otherwise, percentage gains denote relative error reduction from Tucker to NTD-PL. The main real-data evidence is based on all CAVE scenes under matched Tucker rank; noisy reconstruction and cross-domain diagnostics are used to test robustness and operating conditions.

A. Controlled Nonlinear Tensor Recovery

The controlled experiment isolates the measurement-response mechanism without missing entries or observation noise. We start from a latent Tucker tensor $\mathcal{S}^\star$ of known rank and generate the measured tensor by adding a nonlinear response component that is orthogonal to the constant and linear channels. Let $\bar{\mathcal{S}}$ denote the normalized latent tensor and let $\bar{\mathcal{R}}$ denote the normalized nonlinear residual after this projection. The clean measurement is generated as

$$\mathcal{Y}_\alpha = \text{scale}_{[0,1]} \left(\sqrt{1-\alpha}\, \bar{\mathcal{S}} + \sqrt{\alpha}\, \bar{\mathcal{R}} \right), \tag{32}$$

where α controls nonlinear residual energy after the constant and linear response components have been removed. This construction keeps the intended latent rank fixed while making the measured tensor increasingly mismatched to a linear Tucker observation model.

Figure 1 shows the expected transition. When α is small, the measured tensor is close to the latent Tucker signal and NTD-PL stays close to matched-rank Tucker. As the nonlinear residual energy increases, the scalar link explains a growing part of the residual and the gap widens across all four target links: s^2, s^2+s^3, $\tanh(\kappa s)$, and $(e^{\kappa s}-1)/\kappa$. The degree-dependent behavior is reported in Appendix C-A.

All experiments in the main text use the power dictionary because it is compact, interpretable, and directly supports the rank-lift analysis in Section V. Appendix C-A adds controlled details, including a linear-response case in which high-order channels remain inactive. These full-observation experiments serve as mechanism checks; masked recovery and real tensor recovery are evaluated in the following subsections.

B. Hyperspectral Tensor Recovery

We next evaluate whether the same linked Tucker representation improves real spectral-spatial tensors. Unless otherwise stated, hyperspectral cubes are globally normalized, the Tucker backbone rank is fixed across Tucker and NTD-PL, and completion metrics are computed only on held-out entries. We report root mean square error (RMSE) and spectral angle mapper (SAM); lower values are better for both.

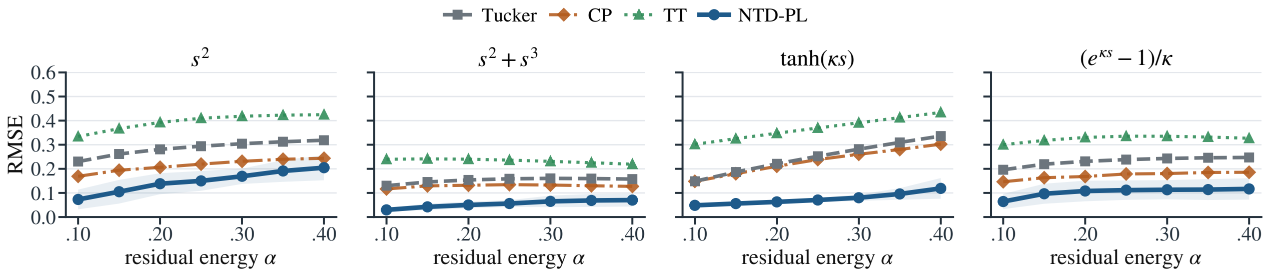

Fig. 1. Controlled nonlinear residual sweep for four target links. The parameter α controls nonlinear residual energy after removing constant and linear components. NTD-PL remains close to Tucker in the nearly linear regime and separates as the nonlinear component grows.

1) CAVE Reconstruction and Completion: We first use the CAVE multispectral image database [23], [24], which provides full spectral-spatial scenes and therefore supports both full-observation reconstruction and masked completion. Full reconstruction tests whether the link adds compact representation power at a fixed backbone rank. Across the tested ranks, NTD-PL consistently improves matched-rank Tucker in both RMSE and SAM when averaged over all 15 scenes, with larger gains at lower ranks and smaller gains as the Tucker rank grows; the full capacity curves are reported in Appendix C-C. This is the expected behavior of a fixed-backbone enhancement: if ordinary multilinear rank is already high enough to absorb the nonlinear response, the marginal value of the scalar link should decrease.

Table I asks a complementary capacity question: how much ordinary Tucker rank is needed to match a lower-rank NTD-PL model? Across the tested ranks, Tucker needs roughly 51–60% more parameters to match the NTD-PL RMSE, while NTD-PL still has lower SAM. The result supports the rank-lift interpretation in Section V: the power link reuses a shared low-rank backbone to express structured nonlinear residuals with a small number of scalar response parameters.

We then add observation noise only to the full-reconstruction setting. The model is fitted to clean or noisy full CAVE cubes and evaluated against the corresponding clean cube. This separates radiometric robustness from missing entry recovery. Across clean observations and Gaussian noise levels down to 10 dB, NTD-PL maintains positive RMSE and SAM gains over matched-rank Tucker; the full table is reported in Appendix C-C.

Held-out completion tests whether the representation benefit transfers to unobserved entries without introducing observation noise. Table II reports CAVE completion under random masks using all 15 scenes and three mask seeds per scene. At the same backbone rank, NTD-PL improves missing-entry RMSE by 12.5–17.7% and missing-entry SAM by 13.4–29.3% across the tested missing rates. The gains are stable across mask density, and paired scene-level summaries are reported in Appendix C-C.

C. Diagnostics and Operating Regimes

The diagnostic D_{link} is computed by fitting a Tucker backbone and measuring the residual reduction obtained by a single frozen-backbone Link Refresh step. It is therefore substantially cheaper than full NTD-PL training and is used here as a model-selection signal for the proposed recovery model. Table III groups CAVE completion scenes by diagnostic tercile. Higher D_{link} corresponds to larger held-out RMSE gains at every missing rate, while the low-diagnostic group has little or negative average gain. The Spearman correlations, 0.829–0.889, show that the diagnostic is strongly aligned with the eventual benefit of the full model.

Figure 2 evaluates the same diagnostic beyond hyperspectral data. Each point is an independently fitted tensor unit, and the horizontal axis is D_{link} measured before full NTD-PL training. Object-view tensors show the clearest separation, action-video tensors follow the same trend, and natural images occupy a lower-yield but still positive regime. These results reinforce the main interpretation: NTD-PL is most effective when the Tucker residual contains shared scalar response structure.

VII. Conclusion

This paper developed dictionary-linked Tucker models for nonlinear low-rank tensor signal recovery. The central idea is to keep the Tucker rank budget tied to the latent signal while modeling the measured tensor through a learned entrywise response. The power-dictionary instance produces a compact shared-backbone rank lift: polynomial channels can increase the effective rank of the observation without introducing an independent high-rank correction. Other finite scalar dictionaries can be used in the same formulation when application-specific response priors are available, but the paper uses the power dictionary as the default link.

The analysis formalized this mechanism through rank expansion, response generation, scalar-response approximation, and nonconvex descent results. The estimator combines a low-dimensional ridge update for the link with gradient-based Tucker updates through the current nonlinear response, and the frozen-backbone diagnostic checks whether a signal is likely to benefit from the link before full nonlinear training.

Experiments support the operating regime of the model. Controlled tensors verify that gains appear when the residual is nonlinear and the active polynomial degree is expressive enough. Hyperspectral reconstruction and completion show that, at matched Tucker rank, the link can improve both RMSE and spectral angle metrics, and rank-lift comparisons show that similar RMSE can require substantially larger ordinary Tucker models. Noisy-observation experiments show that the improvement remains positive under perturbed measurements.

TABLE I
RANK-LIFTED TUCKER COMPARISON ON CAVE RECONSTRUCTION. FOR EACH NTD-PL BACKBONE RANK, WE REPORT THE FIRST HIGHER-RANK TUCKER MODEL MATCHING NTD-PL RMSE AND THE ADDITIONAL PARAMETER COST.

NTD-PL reference				First Tucker matching RMSE				Extra params
Rank	Params	RMSE↓	SAM↓	Rank	Params	RMSE↓	SAM↓	
$(18, 18, 3)$	20k	0.0319	**19.23**	$(28, 28, 3)$	31k	**0.0316**	21.41	$+59.5\%$
$(24, 24, 4)$	27k	0.0256	**15.15**	$(35, 35, 4)$	41k	**0.0254**	16.26	$+51.3\%$
$(33, 33, 4)$	38k	0.0223	**14.41**	$(49, 49, 4)$	60k	**0.0222**	15.28	$+56.5\%$

TABLE II
CAVE RANDOM-MISSING COMPLETION AT MATCHED TUCKER RANK. METRICS ARE EVALUATED ONLY ON HELD-OUT ENTRIES; GAINS ARE RELATIVE IMPROVEMENTS OVER MATCHED-RANK TUCKER.

Missing rate	RMSE*↓			SAM*↓		
	Tucker	NTD-PL	Rel. gain	Tucker	NTD-PL	Rel. gain
0.1	0.0263 ± 0.0179	$\mathbf{0.0229 \pm 0.0158}$	12.9%	11.82 ± 5.33	$\mathbf{8.36 \pm 3.38}$	29.3%
0.3	0.0263 ± 0.0179	$\mathbf{0.0230 \pm 0.0158}$	12.5%	14.75 ± 6.33	$\mathbf{12.47 \pm 5.06}$	15.5%
0.5	0.0264 ± 0.0179	$\mathbf{0.0230 \pm 0.0158}$	12.9%	15.56 ± 6.85	$\mathbf{13.48 \pm 5.75}$	13.4%
0.7	0.0266 ± 0.0180	$\mathbf{0.0219 \pm 0.0128}$	17.7%	16.02 ± 7.21	$\mathbf{13.79 \pm 5.43}$	13.9%

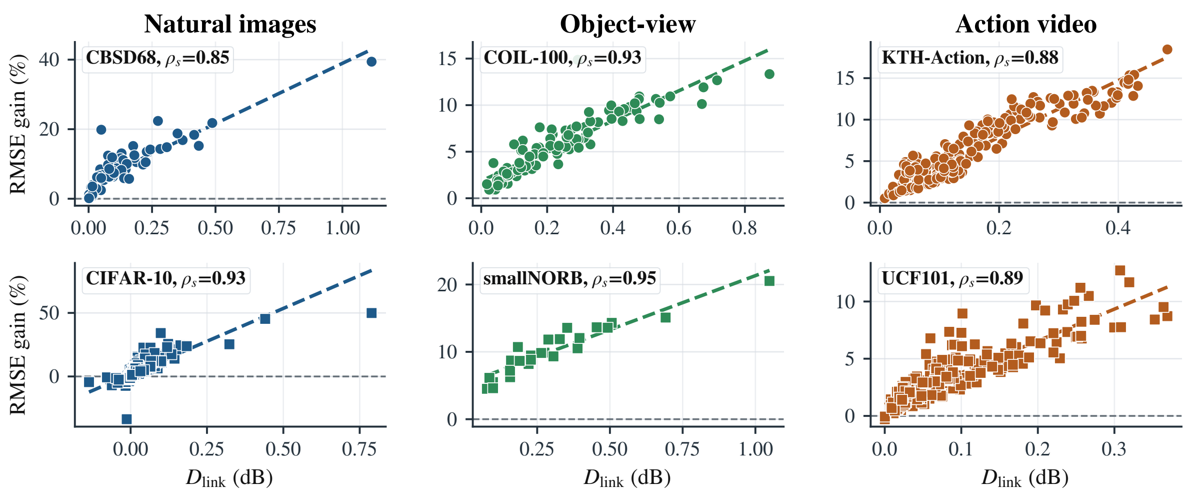

Fig. 2. Cross-domain link diagnostic. Each point plots the frozen-backbone diagnostic D_{link} against the RMSE gain of full NTD-PL over Tucker. Larger diagnostic yield predicts larger benefit from the scalar link.

TABLE III
CAVE COMPLETION DIAGNOSTIC. SCENES ARE GROUPED BY D_{link} TERCILE; ENTRIES ARE MEAN NTD-PL RMSE GAINS OVER MATCHED-RANK TUCKER.

Missing rate	Low D_{link}	Mid D_{link}	High D_{link}	Spearman ρ_s
0.1	0.75%	7.92%	12.63%	0.829
0.3	-1.13%	9.10%	13.12%	0.871
0.5	-1.09%	9.16%	13.18%	0.889
0.7	-0.62%	9.22%	13.07%	0.868

Diagnostic stratification further clarifies the operating regime: when the residual is not well explained by a shared scalar response, increasing rank or changing the tensor prior is more appropriate.

Several directions remain open. Monotone splines or physics-informed response families may be preferable when calibration constraints are known. Sharper sample-complexity or identifiability results, richer noise models, and domain-specific response priors are also natural extensions of the framework.

APPENDIX A
ADDITIONAL MODEL DETAILS

A. Explicit p-Fold Tucker Motivation

The power-dictionary channel in (6) can be viewed as a tied version of an explicit high-order Tucker interaction. This construction is useful because it separates two ideas: the high-order interaction pattern that a polynomial channel represents, and the parameter sharing that makes the proposed model compact.

For a Tucker backbone, a formal p-fold Tucker channel may be written as

$$\widetilde{Y}_{\mathbf{i}}^{(p)} = \sum_{\mathbf{r}^{(1)},\ldots,\mathbf{r}^{(p)}} \left(\prod_{q=1}^{p} X_{r_1^{(q)},\ldots,r_N^{(q)}} \right) \prod_{n=1}^{N} \mathcal{A}^{(n,p)}_{i_n, r_n^{(1)},\ldots,r_n^{(p)}}. \tag{33}$$

For $p = 1$, this is the ordinary Tucker entry formula. For $p = 2$, each mode projection couples two latent Tucker index tuples, giving a quadratic Tucker-style channel. More generally, (33) is the tensor analogue of augmenting a linear map with polynomial interactions among latent coordinates.

The explicit construction is not the parameterization used by the estimator: each mode projection $\mathcal{A}^{(n,p)}$ has $I_n R_n^p$ entries and a separate set of such projections would be needed for every active degree. Dictionary-linked Tucker keeps the same interaction pattern but ties each p-fold projection to the original Tucker factor,

$$\mathcal{A}^{(n,p)}_{i_n, r_n^{(1)}, \ldots, r_n^{(p)}} = \prod_{q=1}^{p} A^{(n)}_{i_n, r_n^{(q)}}. \tag{34}$$

Substituting (34) into (33) gives exactly $\mathcal{S}^{\odot p}$, where $\mathcal{S} = [\![\mathcal{X}; \mathbf{A}^{(1)}, \ldots, \mathbf{A}^{(N)}]\!]$. Thus a degree-$p$ power-link channel is a shared-projection p-fold Tucker interaction: it couples p latent Tucker tuples while reusing the same mode factors.

This viewpoint explains why the model can behave like a rank-lifted tensor representation without storing an untied high-order tensor block. The additional trainable variables are the scalar link coefficients, while the high-order features are induced by repeated use of the same backbone.

APPENDIX B
PROOFS

A. Proof of Proposition 1

Let

$$S_{\mathbf{i}} = \sum_{r_1, \ldots, r_N} X_{r_1, \ldots, r_N} \prod_{n=1}^{N} A^{(n)}_{i_n, r_n}.$$

Expanding $S_{\mathbf{i}}^p$ gives products of p latent Tucker tuples. For each mode n, the resulting mode-dependent terms are all degree-p monomials in the row $A^{(n)}_{i_n,:}$. These monomials are indexed by multi-indices $\alpha \in \mathbb{N}^{R_n}$ with $|\alpha| = p$, of which there are $\binom{R_n+p-1}{p}$. Collecting identical monomials yields a Tucker representation of $\mathcal{S}^{\odot p}$ whose mode-n factor is the degree-p Veronese lift of $\mathbf{A}^{(n)}$. This proves the stated rank bound.

B. Proof of Theorem 1

We first record a standard fact about Veronese evaluations. Let $D = \binom{R+p-1}{p}$. There exists a matrix $\mathbf{A} \in \mathbb{R}^{D \times R}$ whose degree-p Veronese feature matrix has rank D. Indeed, the homogeneous degree-p monomials in R variables are linearly independent polynomials, so one can choose D evaluation points for which the square evaluation matrix is nonsingular. Consequently, for $I \geq D$, a generic $\mathbf{A} \in \mathbb{R}^{I \times R}$ has a degree-p Veronese lift with full column rank D.

Choose generic factor matrices $\mathbf{A}^{(n)} = [a_1^{(n)}, \ldots, a_R^{(n)}] \in \mathbb{R}^{I_n \times R}$ and define the CP-diagonal Tucker backbone

$$\mathcal{S} = \sum_{r=1}^{R} a_r^{(1)} \otimes \cdots \otimes a_r^{(N)}.$$

For generic factors, each mode unfolding of $\mathcal{S}$ has rank R, so the backbone has exact Tucker rank $(R, \ldots, R)$.

Now set the degree-p link coefficient to one and all other coefficients to zero, so $\mathcal{Y} = \mathcal{S}^{\odot p}$. Expanding entrywise gives

$$Y_{\mathbf{i}} = \left(\sum_{r=1}^{R} \prod_{n=1}^{N} A^{(n)}_{i_n, r} \right)^p = \sum_{|\alpha|=p} c_\alpha \prod_{n=1}^{N} \prod_{r=1}^{R} \left(A^{(n)}_{i_n, r} \right)^{\alpha_r},$$

where $c_\alpha = p!/(\alpha_1! \cdots \alpha_R!) > 0$. Let $\mathbf{B}^{(n)} = \mathbf{A}^{(n)\langle p \rangle} \in \mathbb{R}^{I_n \times D}$ be the Veronese lift. By the preceding full-rank fact, each $\mathbf{B}^{(n)}$ has column rank D for generic $\mathbf{A}^{(n)}$. The mode-n unfolding of $\mathcal{Y}$ factors as $\mathbf{B}^{(n)}$ times a full-column-rank right factor built from the remaining lifted factors, with a nonsingular diagonal scaling by the positive coefficients c_α. Therefore every mode unfolding has rank D. The exact standard Tucker rank of $\mathcal{Y}$ is $(D, \ldots, D)$, while $\mathcal{Y}$ is represented by the degree-p linked model using the rank-$(R, \ldots, R)$ backbone.

C. Proof of Corollary 1

By Proposition 1, the degree-p channel can be written as a Tucker tensor with ranks at most $D_n(p) = \binom{R_n+p-1}{p}$. A standalone Tucker parameterization has $\prod_n D_n(p)$ core entries and $\sum_n I_n D_n(p)$ factor entries. In contrast, the power-dictionary linked model reuses the latent core and factors of $\mathcal{S}$ and adds only one scalar coefficient per channel. Summing degrees up to Q adds $Q+1$ link coefficients, proving the stated comparison.

D. Proof of Theorem 2

If $g(s) = \sum_{p=0}^{P} \gamma_p s^p$ and $\mathcal{Y} = g(\mathcal{S}^\star)$, choose the Tucker backbone of the power-dictionary linked model to be the Tucker representation of $\mathcal{S}^\star$ and set $\beta_p = \gamma_p$ for all $p = 0, \ldots, P$. Then $f_{\boldsymbol{\beta}}(\mathcal{S}^\star) = g(\mathcal{S}^\star) = \mathcal{Y}$, giving exact representation with the same backbone rank.

E. Proof of Proposition 2

By the Weierstrass approximation theorem, for every $\epsilon > 0$ there exists a polynomial $q(s) = \sum_{p=0}^{P} \beta_p s^p$ such that $\sup_{s \in [a,b]} |f(s) - q(s)| \leq \epsilon$. Every entry of $\mathcal{S}^\star$ lies in $[a, b]$, so each entrywise approximation error is bounded by ϵ. Therefore

$$\|f(\mathcal{S}^\star) - q(\mathcal{S}^\star)\|_F^2 = \sum_{\mathbf{i}} |f(S^\star_{\mathbf{i}}) - q(S^\star_{\mathbf{i}})|^2 \leq M\epsilon^2,$$

which gives the stated bound after taking square roots.

F. Proof of Proposition 3

For a fixed Tucker backbone, Link Refresh solves the $\boldsymbol{\beta}$-subproblem exactly, hence it cannot increase $\mathcal{F}_Q$. The subsequent backbone update is assumed to satisfy the sufficient decrease condition (21), and therefore also cannot increase $\mathcal{F}_Q$. Combining the two stages gives monotone decrease over the outer iteration. Since $\mathcal{F}_Q$ is bounded below by zero when the regularization weights are nonnegative, summing (21) over k gives

$$c \sum_{k=0}^{K} \left\| \theta^{k+1} - \theta^k \right\|_2^2 \leq \mathcal{F}_Q(\theta^0) - \mathcal{F}_Q(\theta^{K+1}) \leq \mathcal{F}_Q(\theta^0).$$

Letting $K \to \infty$ proves square summability of the steps.

G. Proof of Theorem 3

For fixed Q, the objective is a polynomial function of the Tucker parameters and link coefficients when the monomial basis is used. For any other fixed polynomial basis spanning the same space, it remains polynomial after a linear change of coefficients. The linear-spline dictionary is piecewise polynomial and semi-algebraic; standard RBF dictionaries are analytic and belong to the standard tame classes covered by KL convergence theory. With standard box or normalization constraints used to fix the Tucker gauge, the corresponding constrained objective satisfies the Kurdyka–Lojasiewicz property. Assumption 1 gives sufficient decrease and a relative error bound. The standard KL convergence theorem for nonconvex descent sequences then implies finite length of the iterates and convergence to a critical point of $\mathcal{F}_Q$.

H. Proof of Theorem 4

Fix $0 < \varepsilon \le B_\Theta$, and let $\mathcal{N}_\varepsilon$ be an ε-net of Θ in Euclidean norm. Since Θ lies in a d-dimensional ball of radius B_Θ, it admits a net with

$$|\mathcal{N}_\varepsilon| \le \left(\frac{3B_\Theta}{\varepsilon}\right)^d.$$

For any fixed $\theta_0 \in \mathcal{N}_\varepsilon$, the random variables $\ell_{\mathbf{i}_t}(\theta_0)$ are independent and bounded in $[0, B_\ell]$. Hoeffding's inequality gives

$$\Pr\left(\left|\mathcal{L}(\theta_0) - \widehat{\mathcal{L}}_m(\theta_0)\right| \ge u\right) \le 2\exp\left(-\frac{2mu^2}{B_\ell^2}\right).$$

A union bound over $\mathcal{N}_\varepsilon$ shows that, with probability at least $1-\delta$,

$$\max_{\theta_0\in\mathcal{N}_\varepsilon}\left|\mathcal{L}(\theta_0) - \widehat{\mathcal{L}}_m(\theta_0)\right| \le B_\ell\sqrt{\frac{d\log(3B_\Theta/\varepsilon)+\log(2/\delta)}{2m}}.$$

For an arbitrary $\theta \in \Theta$, choose $\theta_0 \in \mathcal{N}_\varepsilon$ with $\|\theta-\theta_0\|_2 \le \varepsilon$. The uniform Lipschitz assumption implies

$$|\mathcal{L}(\theta) - \mathcal{L}(\theta_0)| \le G\varepsilon, \qquad |\widehat{\mathcal{L}}_m(\theta) - \widehat{\mathcal{L}}_m(\theta_0)| \le G\varepsilon.$$

Combining these two approximation errors with the net bound and taking the supremum over Θ gives the stated inequality.

Appendix C
Additional Experimental Details

A. Controlled Nonlinear Tensors

The controlled experiment uses synthetic tensors whose latent signal is exactly generated by a Tucker model. For each random seed, factor matrices are sampled with normalized columns, a small Tucker core is sampled independently, and the latent tensor is rescaled to a fixed numerical range before the nonlinear response is applied. The target response family is

$$h \in \left\{s^2,\ s^2+s^3,\ \tanh(\kappa s),\ (e^{\kappa s}-1)/\kappa\right\}. \tag{35}$$

For each response, the nonlinear component is projected away from the constant and linear span on the sampled latent entries. The resulting residual is then normalized so that α controls nonlinear residual energy after removing the constant and linear response components:

$$\mathcal{Y}_\alpha = \mathrm{scale}_{[0,1]}\left(\sqrt{1-\alpha}\,\bar{\mathcal{S}} + \sqrt{\alpha}\,\bar{\mathcal{R}}\right). \tag{36}$$

This construction makes the mechanism visible: a linear Tucker model receives the same latent rank budget for every α, while the observation becomes increasingly incompatible with a purely linear measurement map.

All controlled runs are full-observation reconstructions. Tucker, CP, tensor-train, tensor-ring, and NTD-PL are evaluated by RMSE on the generated measurement tensor. NTD-PL uses the same backbone rank as the matched Tucker baseline and varies only the active polynomial degree. The α sweep is visualized in Fig. 1; the supplementary checks below focus on degree-dependent behavior and linear-response consistency under the same controlled generator. Table IV gives a linear-response control: when the measurement is linear and unbiased, the high-order channels remain nearly inactive and NTD-PL reduces to Tucker-level behavior; when a constant offset is present, freeing β_0 absorbs the bias without requiring high-order terms.

TABLE IV
Controlled linear-response consistency. The column $p > 1$ reports the total contribution of nonlinear polynomial channels.

Method	bias $= 0$		bias $= 0.5$	
	RMSE↓	$p > 1$	RMSE↓	$p > 1$
Tucker	**0.0114**	–	0.0647	–
NTD-PL, $\beta_0 = 0$	0.0116	4.19%	0.0522	39.17%
NTD-PL, β_0 free	0.0117	3.19%	**0.0118**	2.98%

B. Metrics and Normalization

For a test index set Γ, the RMSE is

$$\mathrm{RMSE} = \left(|\Gamma|^{-1}\sum_{\mathbf{i}\in\Gamma}(Y_\mathbf{i} - \widehat{Y}_\mathbf{i})^2\right)^{1/2}. \tag{37}$$

For completion, Γ contains held-out entries only. For full reconstruction, Γ is the full tensor support. Spectral angle mapper is computed pixelwise on spectral vectors,

$$\mathrm{SAM}(\mathbf{y},\widehat{\mathbf{y}}) = \arccos\frac{\langle\mathbf{y},\widehat{\mathbf{y}}\rangle}{\|\mathbf{y}\|_2\,\|\widehat{\mathbf{y}}\|_2 + \epsilon}, \tag{38}$$

and then averaged over valid pixels. The small ϵ is used only to avoid division by zero for empty or near-empty spectra. Hyperspectral cubes are globally normalized before fitting, and all compared methods use the same normalization and the same train/test masks.

C. Hyperspectral Protocol

The CAVE multispectral image database [23], [24] is used for the main hyperspectral experiments because it provides fully observed spectral-spatial scenes. Full reconstruction evaluates the fitted tensor on all entries; in the noisy reconstruction

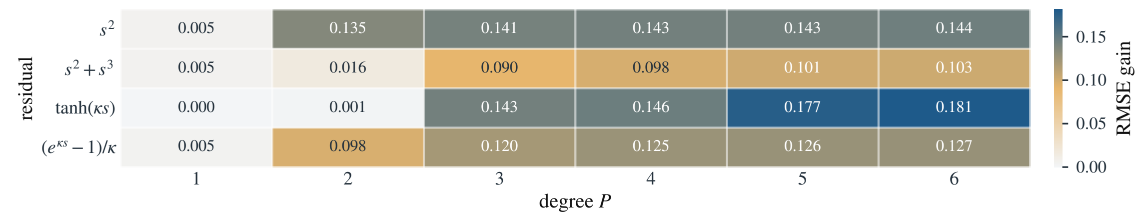

Fig. 3. Degree-dependent controlled gains. Entries report the RMSE reduction of NTD-PL relative to matched-rank Tucker; gains appear when the active polynomial degree can approximate the response.

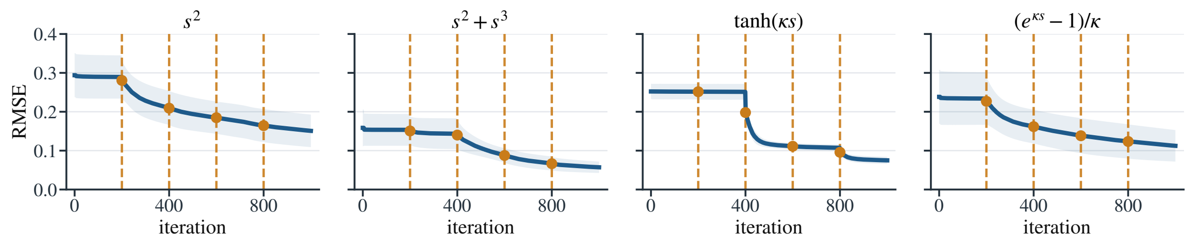

Fig. 4. Representative optimization traces in the controlled experiment. Degree continuation activates nonlinear channels gradually, so the model starts from the Tucker-compatible linear response and then refines the shared scalar link.

TABLE V
SCENE-LEVEL CONSISTENCY ON CAVE FULL RECONSTRUCTION UNDER CLEAN AND NOISY OBSERVATIONS. W/L COUNTS SCENES WHERE NTD-PL IMPROVES OR WORSENS MATCHED-RANK TUCKER; GAINS ARE ABSOLUTE METRIC REDUCTIONS AGAINST THE CLEAN TENSOR.

SNR	RMSE gain			SAM gain		
	W/L	Median	Wilcoxon p	W/L	Median	Wilcoxon p
Clean	15/0	0.0011	3.05×10^{-5}	12/3	0.3399	0.008
40 dB	15/0	0.0011	3.05×10^{-5}	12/3	0.3415	0.008
30 dB	15/0	0.0011	3.05×10^{-5}	12/3	0.3416	0.008
20 dB	15/0	0.0011	3.05×10^{-5}	12/3	0.3598	0.006
10 dB	15/0	0.0012	3.05×10^{-5}	14/1	0.9096	7.63×10^{-4}

TABLE VI
SCENE-LEVEL CONSISTENCY ON CLEAN CAVE RANDOM-MISSING COMPLETION. W/L COUNTS SCENES WHERE NTD-PL IMPROVES OR WORSENS MATCHED-RANK TUCKER AFTER AVERAGING MASK SEEDS; GAINS ARE ABSOLUTE METRIC REDUCTIONS ON HELD-OUT ENTRIES.

Missing rate	RMSE gain			SAM gain		
	W/L	Median	Wilcoxon p	W/L	Median	Wilcoxon p
0.1	12/3	0.0014	0.003	13/2	1.7338	8.54×10^{-4}
0.3	12/3	0.0014	0.003	11/4	0.6953	0.041
0.5	12/3	0.0014	0.003	10/5	0.4144	0.151
0.7	12/3	0.0014	0.002	11/4	0.3021	0.151

check, Gaussian noise is added to the full observation used for fitting and metrics are computed against the clean cube. Random-missing completion samples clean observation masks at the requested missing rates and evaluates RMSE and SAM only on the held-out entries. Tucker and NTD-PL use identical backbone ranks; the default NTD-PL instance adds only the power-link coefficients and the associated training schedule. Table V reports scene-level consistency for full reconstruction at the main matched rank, and Table VI reports the corresponding paired consistency summary for clean random-missing completion.

The noisy-observation robustness evaluation uses the full-reconstruction protocol on all 15 CAVE scenes resized to 128×128 spatial resolution with rank $(12, 12, 6)$. Gaussian noise is added after global max normalization at SNR levels 40, 30, 20, and 10 dB; the clean case uses the same protocol without added noise. The clean evaluation tensor is shared by Tucker and NTD-PL for each scene. Both methods use a shorter iteration budget than the main CAVE experiments, making this a controlled noise-robustness check complementary to the full-resolution clean reconstruction and completion benchmarks.

External hyperspectral experiments use Jasper Ridge, Samson, and Urban cubes after the same global normalization. Table VIII reports the shapes and backbone ranks used in the external reconstruction check, and Table IX reports the corresponding full-reconstruction results. The same fixed-backbone NTD-PL model improves RMSE by 3.6–7.9% and also improves SAM on all three cubes. The NMSE changes are positive as well, with gains of 0.32–0.72 dB. These results serve as a supplementary cross-dataset check beyond the CAVE acquisition setting. Additional low-diagnostic checks verify the expected operating regime: when the frozen-backbone diagnostic is near zero, the scalar link is not expected to provide a reliable improvement.

The rank-lift comparison in Table I searches over higher Tucker ranks and reports the first Tucker model whose RMSE matches the lower-rank NTD-PL reference. Its purpose is to quantify the parameter increase required by ordinary Tucker rank expansion when the residual is link-like.

D. Link Diagnostic Details

The diagnostic D_{link} is computed from a frozen Tucker fit. First, a matched-rank Tucker model is trained under the same mask and normalization as the main experiment. Second, the latent Tucker reconstruction is held fixed and the scalar link coefficients are refit by the ridge update in (11). The diagnostic is the fractional residual energy removed by this one link

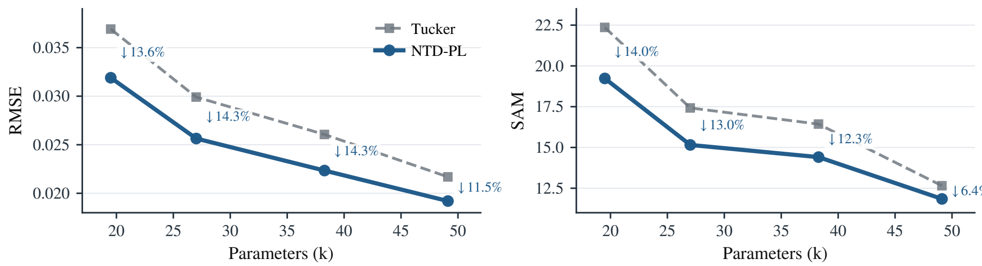

Fig. 5. CAVE full reconstruction at matched Tucker backbone rank. NTD-PL improves both RMSE and SAM, and the gap decreases as Tucker rank increases.

TABLE VII
CAVE NOISY-OBSERVATION RECONSTRUCTION. MODELS ARE FITTED TO CLEAN OR NOISY FULL OBSERVATIONS, AND METRICS ARE COMPUTED AGAINST THE CLEAN TENSOR; GAINS ARE RELATIVE ERROR REDUCTIONS OVER MATCHED-RANK TUCKER.

SNR	Clean RMSE↓			Clean SAM↓		
	Tucker	NTD-PL	Rel. gain	Tucker	NTD-PL	Rel. gain
Clean	0.0320 ± 0.0274	**0.0305 ± 0.0266**	4.86%	14.71 ± 7.57	**13.87 ± 6.73**	5.74%
40 dB	0.0320 ± 0.0274	**0.0305 ± 0.0266**	4.86%	14.72 ± 7.58	**13.87 ± 6.73**	5.77%
30 dB	0.0320 ± 0.0274	**0.0305 ± 0.0266**	4.86%	14.71 ± 7.55	**13.87 ± 6.73**	5.73%
20 dB	0.0321 ± 0.0274	**0.0305 ± 0.0266**	4.87%	14.95 ± 7.72	**13.99 ± 6.80**	6.41%
10 dB	0.0326 ± 0.0273	**0.0310 ± 0.0265**	4.88%	16.01 ± 7.93	**14.63 ± 6.86**	8.58%

TABLE VIII
EXTERNAL HYPERSPECTRAL RECONSTRUCTION PROTOCOL. TUCKER AND NTD-PL USE THE SAME BACKBONE RANK ON EACH CUBE.

Dataset	Shape	Normalization	Backbone rank
Jasper Ridge	$100 \times 100 \times 198$	global max	$(11, 11, 43)$
Samson	$95 \times 95 \times 156$	global max	$(10, 10, 35)$
Urban	$307 \times 307 \times 162$	global max	$(43, 43, 47)$

TABLE IX
EXTERNAL HYPERSPECTRAL RECONSTRUCTION. NTD-PL USES THE SAME TUCKER BACKBONE RANK AS THE TUCKER BASELINE AND ADDS ONLY THE SCALAR POLYNOMIAL LINK.

Dataset	RMSE↓			SAM↓		
	Tucker	NTD-PL	Rel. gain	Tucker	NTD-PL	Rel. gain
Jasper Ridge	0.0462	**0.0430**	6.9%	9.14	**8.16**	10.7%
Samson	0.0263	**0.0242**	7.9%	5.70	**5.65**	1.0%
Urban	0.0444	**0.0428**	3.6%	8.23	**7.94**	3.5%

update, as defined in (17). This procedure is cheaper than full NTD-PL training because it does not update the Tucker backbone through the nonlinear response.

The CAVE diagnostic table in the main text groups scenes into terciles by $D_{\rm link}$ and compares the eventual held-out RMSE gain of full NTD-PL. We use the same frozen-backbone computation for external checks and low-diagnostic regimes where the link is not expected to help.

E. Cross-Domain Diagnostic Data

The cross-domain diagnostic in Fig. 2 treats each tensor unit as an independent recovery problem. Natural image units include CBSD68 and CIFAR-10 images [25], [26]; object-view units include COIL-100 and smallNORB tensors [27], [28]. Each unit is normalized independently, fitted by matched-rank Tucker and NTD-PL, and summarized by frozen-backbone diagnostic yield versus the eventual RMSE gain. These experiments are diagnostic tensor-recovery checks. Their purpose is to test whether the scalar-response residual pattern appears beyond hyperspectral cubes.

CBSD68 units are resized natural images with rank $(32, 32, 3)$. CIFAR-10 uses 1000 class-balanced test images of shape $32 \times 32 \times 3$ with rank $(20, 20, 3)$. COIL-100 uses one object at a time as a view tensor with rank $(8, 12, 12, 2)$. smallNORB uses one object instance at a time as a stereo view tensor with rank $(12, 24, 24, 2)$. KTH-Action [29] and UCF101 [30] provide the action-video tensor units. All units are normalized independently, and diagnostic labels are assigned from $D_{\rm link}$, not from the eventual NTD-PL gain.

F. Compute Resources

The reported runs used a CPU Python implementation based on NumPy, SciPy, and TensorLy on a workstation with a 13th Gen Intel Core i5-13400F CPU, 32 GB system RAM, Windows 11, and Python 3.13. An NVIDIA RTX 5060 GPU with 8 GB memory was present, but the reported scripts do not use GPU kernels. Sweep scripts cap BLAS/OpenMP threads per worker and parallelize independent jobs, using up to 12 workers for the main low-rank sweeps and 6 workers for the diagnostic batches.

From the recorded runtime fields, controlled synthetic residual jobs were sub-second to about 1 s per configuration. Full-resolution CAVE reconstruction took about 10–17 s per scene for matched-rank Tucker and 172–217 s per scene for NTD-PL in the main low-rank reconstruction sweep. CAVE random completion runs took about 0.9–18.6 s per scene/mask/seed, while structured completion runs were about 1.6–5.0 s. The frozen-backbone CAVE link diagnostic itself took about 0.12–0.59 s per fitted refresh. Non-HSI diagnostic batches with 6 CPU workers took about 77 s for CBSD68/CIFAR-10/COIL-100/smallNORB and about 5–6 minutes for each action-video

TABLE X
DATASET-LEVEL STATISTICS BEHIND THE CROSS-DOMAIN DIAGNOSTIC SCATTER IN FIG. 2. GAIN COLUMNS REPORT MEAN NTD-PL RMSE IMPROVEMENT OVER INDEPENDENTLY FITTED MATCHED-RANK TUCKER WITHIN D_{link} TERCILES.

Domain	Dataset	Units	Med. D_{link}	ρ_s	Low	Mid	High
Natural images	CBSD68	68	0.091	0.85	4.17%	8.40%	14.60%
Natural images	CIFAR-10	1000	0.008	0.93	-0.01%	2.59%	7.85%
Object-view	COIL-100	100	0.208	0.93	2.77%	5.46%	9.14%
Object-view	smallNORB	25	0.264	0.95	6.92%	9.91%	14.16%
Action video	KTH-Action	240	0.107	0.88	2.74%	4.27%	10.87%
Action video	UCF101	240	0.065	0.89	1.44%	2.97%	5.89%

batch. Peak memory was not instrumented per run; empirically the largest reported CPU runs completed within the 32 GB RAM budget.

G. Dataset Asset and Usage Information

All datasets used in the experiments are public research benchmarks. They are used only for reconstruction, completion, or diagnostic evaluation; processed summaries and scripts are redistributed rather than raw dataset files.

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TABLE XI
DATASET SOURCES AND USAGE NOTES. "NO" IN THE REDISTRIBUTION COLUMN MEANS THAT RAW DATA FILES ARE NOT INCLUDED.

Dataset	Source / citation	License or usage note	Redistribution / use
CAVE Multispectral Image Database	Columbia CAVE Laboratory; [23], [24]	Public download and citation information are provided by the official page; used only for research evaluation.	No; reconstruction, completion, diagnostics.
CBSD68 / Berkeley natural images	Berkeley Segmentation Dataset; [25]	Official usage statement permits research and non-commercial use and requests citation.	No; non-HSI diagnostic evaluation.
CIFAR-10	University of Toronto CIFAR page; [26]	Public research benchmark download and citation information are provided by the official page.	No; non-HSI diagnostic evaluation.
COIL-100	Columbia Object Image Library; [27]	Public download and citation information are provided by the official page; used only for research evaluation.	No; non-HSI diagnostic evaluation.
smallNORB	NYU/Yann LeCun dataset page; [28]	Official page provides the database for research purposes and requests citation.	No; non-HSI diagnostic evaluation.
KTH-Action	KTH action-recognition dataset page; [29]	Official page states that the database is publicly available for non-commercial use and requests citation.	No; non-HSI diagnostic evaluation.
UCF101	UCF CRCV dataset page; [30]	Public benchmark download and citation information are provided by the official page; used only for research evaluation.	No; non-HSI diagnostic evaluation.

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